

Snowflake Load

Session 03 · ISBA 4715 · MP02

The **L** in ELT — move Basket Craft raw tables from AWS RDS into a cloud data warehouse.

PostgreSQL saves orders; Snowflake scans millions



PostgreSQL

OLTP · row-oriented
one order at a time



Snowflake

OLAP · columnar
millions of rows per query

Different databases for different jobs — operational versus analytical.

Storage and compute are separate



**The idea that explains everything
else about Snowflake.**

Storage: cheap bytes, always on

column: customer_id

column: order_date

column: order_value

- Columnar files in cloud storage
- Pay for bytes stored
- Always available, low cost

Compute bills per second, suspends when idle



- Cluster that runs SQL
- Pay per second running
- Zero cost when suspended
- Resize with one setting

Separation lets cost and speed scale independently

XS warehouse

BI

L warehouse

dbt

Suspended

\$0

Shared storage · always on · never resized

- Auto-suspend stops the meter
- Resize up, drop back down
- dbt jobs stay free-tier fast

Today you only do the "L"

ELT

- Create warehouse, database, `raw` schema
- Copy RDS tables into `raw`
- No cleaning, no transforms
- Session 04 builds the star schema

In industry, managed tools land raw data



- Click-configure a source
- Scheduled loads, schema drift, retries
- Priced by row volume
- Custom Python loaders are rare in production

Writing it yourself teaches what Fivetran hides



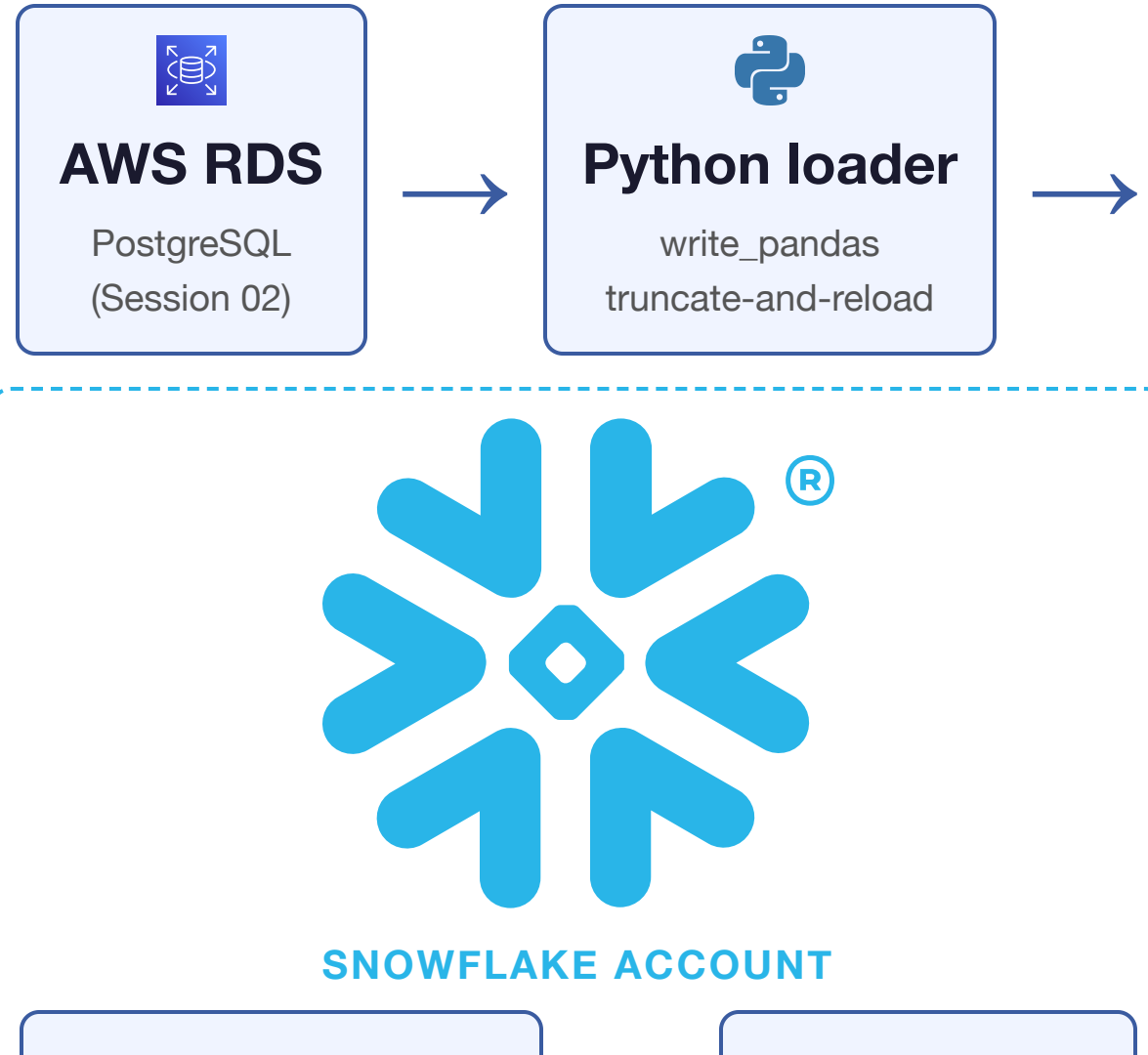
wrapper · opaque
one "connect" button

Your loader

write_pandas · COPY INTO
every building block visible

- Debug Fivetran when it breaks
- Build loaders for systems without connectors
- Know the building blocks, not just the wrapper

RDS → Python → warehouse → raw



Seven steps to a populated raw schema

#	Step	What you do
13	Verify account	Region, edition
14	Create objects	Warehouse, database, <code>raw</code> schema
15	Store credentials	Snowflake vars in <code>.env</code>
16	Brainstorm the loader	Surface design decisions first
17	Implement the loader	Claude Code writes it, official package
18	Run and verify	Row counts match RDS
19	Commit and push	Update <code>CLAUDE.md</code> , ship

Now open the tutorial and jump to Step 13

Tutorial: `mp02-tutorial.md` → [Step 13](#)